MINISTRY OF EDUCATION AND TRAINING

# FPT UNIVERSITY

## Capstone Project Document

Project Title

| IAP491_G.. | | |
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- 11/2025 -

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# Acknowledgments

No one achieves something great without the help of those around them, whether directly or indirectly. To complete this capstone project, we were fortunate to receive tremendous support and assistance from our supervisors, lecturers, friends, and family. We would like to dedicate these opening pages to express our sincere gratitude to everyone who has accompanied us throughout this journey.

First and foremost, we would like to express our deepest appreciation to all the lecturers at FPT University Hanoi Campus, particularly those in the Information Technology Department. The valuable knowledge they have imparted, along with their dedicated support throughout the implementation process, has enabled us to complete this capstone project and achieve noteworthy results.

We would like to extend our special gratitude to our supervisor, MSc. Trịnh Tiến Đạt, and our co-supervisor, Dr. Nguyễn Anh Nhật, who have inspired us, provided dedicated guidance, and offered invaluable support in terms of knowledge. They have created a favorable environment for us to learn and exchange ideas with fellow students in the research group. These are precious knowledge and experiences that will benefit us not only in this capstone project but also in our future careers.

Finally, we wish to express our heartfelt thanks to our families and loved ones, who have always been our solid support and have encouraged every decision we have made.

Despite our best efforts to make this thesis as complete as possible, shortcomings and limitations are inevitable. We sincerely hope to receive your understanding and constructive feedback from our supervisors and readers.

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# Abstract

A good abstract is a stand-alone summary of the paper, and should summarize the key components of the manuscript. Generally, the abstract should be concise and informative within 200-300 words. As an abstract is a separate section, it should be a self-containing text (no abbreviations, no references, no URLs, no undefined concepts, etc.). For research articles, abstracts should give a pertinent overview of the work. We strongly encourage authors to use the following style of structured abstracts, but without headings: (1) Background: Place the question addressed in a broad context (research background) and highlight the purpose of the study; (2) Methods: Describe briefly the main methods or treatments applied; (3) Results: Summarize the article's main findings; and (4) Conclusions: Indicate the main conclusions or interpretations.

**Keywords:** cybersecurity, information security, threat analysis, resilience

# List of Abbreviations

| **Acronym** | **Definition** |
|---|---|
| DSP | Digital Service Providers |
| EU | European Union |
| GSP | Global Stability Protection |
| OES | Operators of Essential Services |

# Chapter 1

# Introduction

## 1.1 Background

The Internet of Things (IoT) connects physical objects, sensors, and various devices through communication networks, enabling data-driven services in domains such as healthcare, industrial automation, and smart environments. Alongside the expansion of IoT deployments, the number of connected components exposed to cybersecurity threats increases correspondingly [1]. The Message Queuing Telemetry Transport (MQTT) protocol is commonly used for lightweight publish-subscribe communication in IoT environments. However, MQTT-based systems can become targets for denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks. The DoS/DDoS-MQTT-IoT dataset, collected from a physical IoT testbed, provides normal MQTT traffic alongside ten DoS attack scenarios to facilitate the study of attacks in this context [2]. Intrusion Detection Systems (IDS) monitor network activity to identify traffic or behavior exhibiting malicious signs. Machine learning and deep learning methods have been utilized and researched for this task due to their ability to learn characteristic patterns directly from network traffic. Their effectiveness depends partly on the availability and representativeness of the training data [3]. Federated Learning (FL) allows multiple clients to collaboratively train a shared model while keeping their data local; a central server distributes the global model and aggregates updates from clients across training rounds [4]. This approach is highly suitable when IoT data is distributed across multiple devices or locations, although communication and computational costs remain critical factors to consider in resource-constrained environments [5, 6]. Keeping data local also does not equate to completely eliminating privacy concerns arising from the exchange of model updates [7].

Privacy-preserving synthetic data generation is a potential solution to address the shortage of training data for rare attack classes in IoT networks while still ensuring strict security guar-

antees. Rather than relying on deep neural networks that require substantial computational resources and face convergence difficulties on edge devices [8], the probabilistic graphical model paradigm (Probabilistic Graphical Models) approaches the problem by decomposing tabular data into low-dimensional marginal frequency distribution slices (low-dimensional marginals) [9]. Through the Chow-Liu dependency tree structure (Chow-Liu Tree) [10], this approach effectively captures the core correlations between traffic features and attack labels without being affected by the curse of dimensionality. When combined with Differential Privacy (DP) [9] on contingency tables and consistent graphical inference mechanisms (such as Private-PGM), the system can reconstruct a valid global distribution to generate conditional traffic data, establishing a foundation to enhance the performance of downstream intrusion detection systems (IDS) [11, 12].

## 1.2 Problem Statement

## 1.3 Research Objectives

## 1.4 Significance of the Study

## 1.5 Scope and Limitations

## 1.6 Thesis Structure

# Chapter 2

# Literature Review

Reviews relevant existing research and demonstrates how your thesis fits into the broader academic conversation.

## 2.1 Review of Previous Studies

## 2.2 Summary of the Literature Review

## 2.3 Contribution of Research

# Chapter 3

# Methodology

Describes the research design, data collection methods, and analysis techniques.

## 3.1 Research Design

## 3.2 Data Collection Methods

## 3.3 Sampling Data Analysis Techniques

## 3.4 Limitations of the Methodology

# Chapter 4

# Experimental and Results

Provide a concise and precise description of the experimental results, their interpretation as well as the experimental conclusions that can be drawn. You should discuss the results and how they can be interpreted in perspective of previous studies and of the working hypotheses.

## 4.1 Introduction

## 4.2 Presentation of Data

## 4.3 Analysis of Results

## 4.4 Interpretation of Results

## 4.5 Comparison with Literature

## 4.6 Implications of the Results

# Chapter 5

# Discussion

The findings and their implications should be discussed in the broadest context possible and limitations of the work highlighted.

## 5.1 Restate the Research Problem or Objectives

## 5.2 Summarize Key Findings

# Chapter 6

# Conclusion and Future Work

The 'conclusions' are a key component of the paper. It should complement the 'abstract' and is normally used by experts to value the paper's engineering content. A conclusion is not merely a summary of the main topics covered or a re-statement of your research problem, but a synthesis of key points and, if applicable, where you recommend new areas for future research.

## 6.1 Conclusion

## 6.2 Future Work

# Appendix A

# If Any

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